

Joshua Jang

MSc Student @ ETH Zurich, Switzerland

Email: hyjang@student.ethz.ch

Portfolio: musicofmusix.github.io

Research Interests

Computer Graphics, Computational Photography, Human-Computer Interaction for Creativity

Education

2026-2028 (Expected)	ETH Zurich, Switzerland Master of Science in Computer Science
2020-2026	University of Toronto, Canada Bachelor of Science in Computer Science and Cognitive Science <i>Advisors: Prof. Michael Brown, Prof. Karan Singh</i> Mandatory Military Service, Republic of Korea (2021-2022)

Research Experience

2025.09-2026.05	University of Toronto, Canada Undergraduate Research Assistant @ Dynamic Graphics Project (DGP) Lab <i>Advisors: Prof. Karan Singh, Karran Pandey</i> - Contributed to an interactive, context-aware copy-paste tool for 3D gaussian splat scenes. Paper under review at SIGGRAPH Asia 2026. - Designed and implemented a pipeline lifting 2D DINOv3 feature embeddings into 3D gaussian splats for segmentation and paste compatibility map generation.
2025.01-2025.05	Samsung Electronics, Canada Research Intern @ AI Center - Toronto <i>Advisors: Prof. Michael Brown, Dr. Hakki Karaimer</i> - Developed a compact, interpretable model of photographic film color for emulation from digital images (see Publications). - Implemented utilities for camera image signal processing (ISP) analysis in Samsung's projects, notably a vectorized RAW renderer and a homography-based automated color chart patch extractor. - Built RAW-JPG image datasets across various environments and ISPs, conducting capture, alignment, and post-processing.

Publications

2025	Hyun Jo (Joshua) Jang, Hakki Karaimer, Michael Brown Emulating Emulsion: A Compact Physically-Based Model for Film Colour <i>ACM SIGGRAPH 2025 Posters</i>
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Teaching Experience

- 2025.09-2026.03 **University of Toronto, Canada.**
Undergraduate Advisor @ Computer Science Student Union
- Mentored a small cohort of undergraduates exploring artist-driven research in computer science, guiding them through weekly literature reviews and a poster project on tangible user interfaces.

Exhibitions

- 2024 **Hart House Camera Club 103rd Annual Exhibition**, Toronto, Canada.
William J. Dowkes Award for People Photography, Winner
- 2023 **Interactive Media Exhibition for Children**, Gimpo, South Korea.
Exhibitor, Artist

Creative Projects

- 2026 **Optical Vinyl Record Tracking**
- Developed a low-cost method for manipulating digital audio via physical vinyl records using only an optical mouse; aimed at DJs.
- 2023 **Interactive Virtual Subway Station**
- Designed and developed a stylized 3D subway station, with visuals replicating 1990s Japanese hand-drawn animation through custom-made non-photorealistic rendering (NPR) shaders in Unity (see **Portfolio**).
 - Installed as part of an in-person exhibition, where attendees would be greeted with a train departing towards the next gallery.
- 2022 **3D + 2D Interactive Poster**
- Developed a motion graphic poster in where real-time 3D + 2D rendering and user interactivity were added in Unity (see **Portfolio**).
 - Implemented a fully animated humanoid character featuring facial expressions, real-time shader replacement, and procedurally generated relaxed cubic splines for stylistic effects.
- 2016-Present **Photography, Graphic Design, 3D Art & Frontend Web Development**
- Continuous practice of arts and design for inspiration.

Skills

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| Programming | Python, C/C++, C# |
| Creativity | Photoshop, Illustrator, Lightroom, Unity, Blender, Cinema 4D, HTML, CSS |
| Languages | English (Fluent/C2), Korean (Native) |